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Catalytic Decomposition of H₂S into H₂ and S: Process Design and Optimization for Future Piloting

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Abstract

H₂S is one of the major toxic byproducts of the oil and gas industry and an undesired pollutant in natural gas. Global H₂S generation is estimated to be over 7 million tons per year. The objective is to present a rising technology to convert H₂S into hydrogen and sulfur as an alternative to the Claus process, which yields water and sulfur. The proposed process involves a reactor with a stationary catalyst bed that decomposes H₂S into hydrogen and sulfur in a few seconds. In-depth laboratory studies of H₂S decomposition were supplemented by the DFT quantum modeling. In terms of value generation efficiency, the technology outperforms conventional Claus and the research results reported lately. Using the process modeling software package, a process flow diagram was created. The data obtained from laboratory testing and computational simulations will be used to create a pilot unit to demonstrate the technology's feasibility. This paper presents novel progress in developing and upscaling the process of H₂S decomposition to hydrogen and sulfur using catalyst. Alternative approaches reported up to date face significant challenges in industrial implementation due to low effectiveness, complexity, or lack of experimental data or process design. Results obtained in this work demonstrate and facilitate the feasibility of the proposed method of H₂S conversion.

Introduction

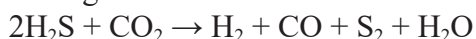
Oil and gas operations account for significant part of hazardous emissions released by industry. Besides just greenhouse gases, inevitable formation of toxic and corrosive gases such as H₂S is also a reality for this industry. It can be found as an undesired pollutant in natural gas and oil wells, or be formed during downstream oil processing. Global H₂S generation is estimated to be over 7 million tons per year. Converting H₂S to hydrogen is an opportunity to create clean hydrogen energy from toxic gases. The Claus process has long been known and used as a method for conversion hydrogen sulfide to sulfur. It has been improved many times since implementation to minimize harmful emissions and increase sulfur recovery, which nowadays exceeds 99%. A rising interest to date is development of a new technology to convert H₂S into hydrogen and sulfur as an alternative to the Claus process, which yields water and sulfur.

Exploring available data sources for hydrogen sulfide (H₂S) to hydrogen (H₂) conversion approaches resulted in a table (Table 1) of technologies which are not yet commercially available. Many of these proposed concepts face challenges which may limit their practical application in commercial-scale applications.

Table 1—H₂S to H₂ technologies located in the study

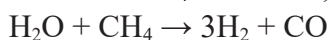
Option #	TRL	Category	Name
1	9	Claus-integrated	SRU 100% O ₂ Enrichment with H ₂ Recovery and CO ₂ Capture / AG2S
2	5	Reformation	H ₂ S Reforming
3	7	Decomposition	Thermal
4	3		Photocatalytic
5	4		Non-thermal Plasma
6	4		Thermochemical
7	6		Electrolysis
8	1		Biological

Options 1 and 2 in the table 1 employ commercially available technologies and could be implemented in the short-term. Option 1 utilizes the fact that during Claus thermal stage H₂S is partially dissociated to H₂ and S at high temperatures. Swapping air for 100% O₂ enriched gas allows to eliminate N₂ from the process gas, resulting in a tail gas stream that is primarily composed of CO₂, H₂O and H₂. H₂ is separated and CO₂ is captured for sequestration or other uses. This option has the added benefit of increasing existing sulfur recovery unit (SRU) capacity by as much as 100%, if required. While this approach presents an opportunity for fairly immediate implementation and provides a potential stepping stone toward a larger strategy of producing H₂ from H₂S, the quantity of H₂ that can be produced by this option is quite low and estimated to be 9 %. Another approach which also falls in option 1 is AG2S process (*Bassani 2017*). Here, the two main pollutants – H₂S and CO₂ – are reduced to syngas in a regenerative thermal reactor according to the following overall reaction:

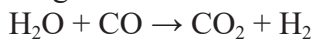


However, such approach does not aim to obtain pure hydrogen as clean fuel, rather as a part of a syngas mixture to further use as a feed in downstream processes, e.g. synthetic fuel synthesis. Separation and purification of H₂ in this instance seems economically impractical.

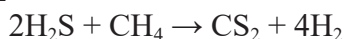
Option 2 is a variation of a well-known process of steam methane reforming (SMR) under high temperature, elevated pressure and in the presence of water vapor. In the steam reforming process, for each molecule of CH₄ converted, 3 molecules of H₂ and 1 molecule of CO are produced:



The carbon monoxide formed reacts with the excess vapor fed to the reformer according to the known water gas shift reaction:



On the other hand, as alternatives to the steam reforming of methane, the pyrolysis processes of CH₄ or H₂S have been evaluated:



Considering H₂S as a feedstock, bearing in mind the current efforts for acid gases emission abatement, hydrogen production via H₂S Methane Reformation (H2SMR) was discussed as an alternative to the traditional steam reforming (*Spatolisano 2022*). Despite high estimated H₂ yield, calculated to be in the

range of 40 to 50 %, the process is very energy consuming, proceeds at high temperature, prone to carbon deposits due to methane pyrolysis. These drawbacks significantly limit the feasibility of such technology.

Options 3 to 8 can be classified as straightforward decomposition of H_2S to H_2 and S using various techniques. These options are extensively explored in scientific literature; their detailed discussion is given elsewhere (*Li 2018, De Crisci 2019, Chan 2023, Palankoev 2024*). The majority of proposed approaches face major obstacles to industrial implementation due to low effectiveness, complexity, or a lack of experimental data or process design. Currently, the thermochemical (option 6) method has the highest commercial potential. Despite a lower estimated TRL compared to other options, this approach provides a reliable and feasible method for H_2S -to- H_2 transformation with high capacity. The "sulfur looping scheme" is a promising design for thermochemical hydrogen sulfide decomposition (*Jangam 2021*). This concept has been demonstrated on a laboratory scale by converting ppm-level concentrations of H_2S in feed. The next step toward industrial process design would be upscaling studies, which would include simulating industrial conditions and assessing potential implementation opportunities, as is the goal of this work.

Description and Application of Equipment and Processes

Catalysts

In this work, several catalysts were synthesized and tested, namely:

- Series of hexaaluminate catalysts: $\text{LaFe}_x\text{Al}_{12-x}\text{O}_{19}$, where $x = 2, 4$ or 6
- Series of iron sulfides (FeS) doped with molybdenum (Mo) or tungsten (W) at a concentration of 1-3 at%
- Ni_3S_2 supported on Al_2O_3

The catalysts were chosen on a basis of literature review as most promising materials. The composition, structure and properties of synthesized materials were verified using XRD, SEM, TEM and BET analysis.

Equipment

3D visualization of the setup used for catalyst testing is depicted in [figure 1](#). The unit consists of gas flow and temperature control module, heated tubular reactor, gas-solid separator, gas analysis module and a data PC for data logging and processing.

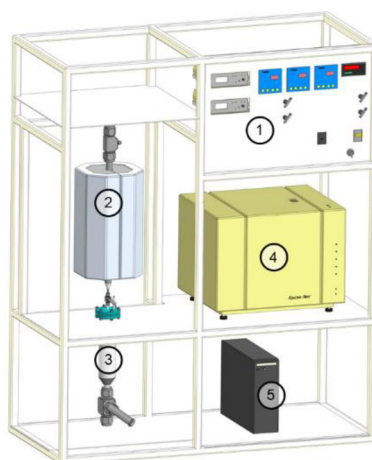


Figure 1—Catalyst testing unit: gas flow and temperature control module (1), heated tubular reactor (2), gas-solid separator (3), gas analysis module (4) and a data PC for data logging and processing (5).

Procedure

Catalytic tests were carried out in a tubular reactor with stationary catalyst bed at a temperature of 500 °C, 10 vol% of H₂S in the feed, and a GHSV (ratio of feed flow in l/h to reactor volume in l) of 1000 h⁻¹. The temperature of 500 °C is the minimum in the studied range of thermocatalytic conversion of hydrogen sulfide and its choice is determined by minimizing energy consumption during the possible implementation of the process, the initial concentration of hydrogen sulfide of 10 vol% was chosen for reasons of modeling a realistically possible flow of acid gas with a high content of H₂S, GHSV equal to 1000 h⁻¹ was selected from the range of typical values for catalytic plants involved in hydrogen sulfide processing (*ZareNezhad 2009*). Analysis of inlet and outlet gases was carried out by gas chromatography. Hydrogen yield was calculated by the equation:

$$Y = \frac{C(H_2)}{C(H_2S)} \times 100\%,$$

where C(H₂) is the concentration of H₂ (vol%), C(H₂S) is the initial concentration of hydrogen sulfide, in this work 10 vol%.

The quantum-chemical calculations were carried out within the density functional theory (DFT), implemented in VASP package. The exchange-correlation functional was calculated in the generalized gradient approximation (GGA) in the Perdue-Burke-Ernzerhof (PBE) parameterization. To describe the ion-electron interaction, the projector augmented wave method (PAW) was used. Integration in the first Brillouin zone was performed using the Monkhorst-Pack method. Relaxation of the atomic structure was carried out until the energy difference between two steps of the ion optimization became less than 10⁻⁴ eV, and the energy difference between two steps of the electron optimization cycle became less than 10⁻⁵ eV. In the calculations, Grimm dispersion corrections were applied to describe the van der Waals interaction (DFT-D3 method). The diffusion barriers were calculated using the nudged elastic band method (NEB). The VESTA programs were used to visualize the atomic structure.

Modelling of heat and mass balances as well as the overall design of an upscaled unit was conducted using conventional chemical engineering process simulator.

Data and Results

Catalysts screening

Comparison of screened catalysts performance is given as an overview in [table 2](#).

Table 2—Performance of screened catalysts in H₂S decomposition to H₂. Operating conditions: 500 °C, 10 vol% of H₂S in the feed (balanced by nitrogen), GHSV of 500-1000 h⁻¹.

Catalyst	H ₂ yield, % (% of H ₂ S converted)	Deactivation
LaFe ₂ Al ₁₀ O ₁₉	1.5	none
LaFe ₄ Al ₈ O ₁₉	1.0	none
LaFe ₆ Al ₆ O ₁₉	1.0	none
Fe _{0.99} Mo _{0.01} S	8.2	Yes, formation of Fe ₇ S ₈
Fe _{0.98} Mo _{0.02} S	8.8	Yes, formation of Fe ₇ S ₈
Fe _{0.97} Mo _{0.03} S	10.0	Yes, formation of Fe ₇ S ₈
Fe _{0.99} W _{0.01} S	5.8	Yes, formation of Fe ₇ S ₈
Fe _{0.98} W _{0.02} S	10.2	Yes, formation of Fe ₇ S ₈
Fe _{0.97} W _{0.03} S	10.5	Yes, formation of Fe ₇ S ₈
Ni ₃ S ₂ supported on Al ₂ O ₃	40.0	Yes, formation of NiS, reversible

As a result of the catalyst screening tests the following can be concluded:

- For $\text{LaFe}_x\text{Al}_{12-x}\text{O}_{19}$ ($x = 2, 4$ or 6) catalysts the H_2 yield is very low under chosen conditions regardless of atomic composition. According to the literature, these catalysts require much higher operating temperature (around 800°C) and much lower H_2S content in the feed (around $0.1\text{ vol}\%$) to perform. On the other hand, no signs of deactivation were observed by XRD analysis;
- Doped iron sulfide catalysts demonstrate intermediate H_2 yield of 10% under chosen conditions. The effect of doping the iron sulfide catalyst with molybdenum or tungsten is observed up to $2\text{ at}\%$, after which it is not beneficial to increase the dopant content regardless of the metal. Deactivation was observed by XRD in the form of the emergence of new Fe_7S_8 phase, which is inactive in H_2S decomposition;
- The best performance is observed for Ni_3S_2 catalyst on alumina support, providing 40% H_2 yield. Under studied operating conditions the catalyst sample maintains its efficiency up to $45\text{--}50$ minutes on stream. As evidenced by XRD analysis, gradual deactivation takes place, while active phase is sulfurized yielding inactive phase: $\text{Ni}_3\text{S}_2 \rightarrow \text{NiS}$. However, this transformation is reversible in a matter of minutes by cutting the H_2S flow and increasing the temperature to 600°C . As a result, the catalyst releases accumulated sulfur restoring active Ni_3S_2 phase, which is confirmed by XRD and TGA analysis and by visual observation of dispersed sulfur particles deposited on the lower and colder parts of the wall of the demonstration reactor (Figure 2).

For more information on laboratory-scale catalyst screening, please see previously published work (Palankoev 2024).



Figure 2—Formation of finely dispersed sulfur particles on the surface of reactor outlet during the catalyst regeneration process.

DFT Modelling

Quantum modeling using the DFT method was conducted to gain a deeper understanding of Ni_3S_2 interactions with sulfur during H_2S decomposition and catalyst regeneration processes. The structure of the Ni_3S_2 crystal surface under consideration is shown in Figure 3. To optimize calculations, some of the sulfur and nickel atoms in the crystal were immobilized (highlighted in the blue area in Figure 3) as their participation in the surface processes is unlikely.

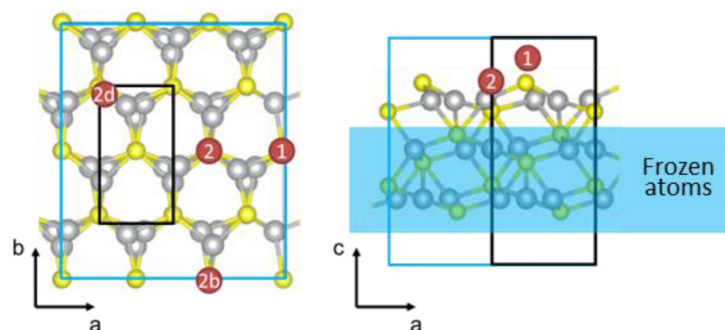


Figure 3—Visualization of the system considered during DFT modelling of H₂S to H₂ decomposition on the Ni₃S₂ crystal. The nickel and sulfur atoms are shown in gray and yellow, respectively. The supercell under consideration is shown by blue solid lines, and the original unit cell is shown by black lines. The "frozen" nickel and sulfur atoms are highlighted in the blue area: the atoms in this area are deemed as not affecting the reaction and are excluded to optimize calculations. Numbers indicate possible H₂S adsorption sites (1 and 2) during reaction phase and possible sites of S₈ molecule migration (2d and 2b) during catalyst regeneration phase.

It was established that on the surface of the Ni₃S₂ crystal there are two nonequivalent sorption positions of single sulfur atoms, designated in Figure 3 5 as positions 1 and 2. The most energetically favorable sorption position is position 2, for which the binding energy is -1.95 eV, while for position 1 the sorption energy is positive and amounts to 0.54 eV. The positive value of the sorption energy indicates that this position is metastable.

To study the process of cleaning the surface of the Ni₃S₂ crystal from sulfur formed during the decomposition reaction of hydrogen sulfide molecules H₂S, the sorption of two closely located sulfur atoms (these sulfur atoms will be designated as S₁ and S₂) on the surface was considered. Initially, two single sulfur atoms were sorbed in positions 1 and 2, and for such a configuration of the system, the sorption energy is -1.39 eV, which means that the sorption of two single sulfur atoms is energetically favorable. Two possible pathways for the process were further considered: (1) the first stage of the splitting reaction was accompanied by the diffusion of one sulfur atom over the surface or (2) the simultaneous diffusion of two sulfur atoms. Evaluation of the energy barriers and energy levels of initial and final stages of the studied system evidences that second process is more favorable. The reaction of S₂ molecule detachment from the surface of the material occurs barrier-free, which is associated only with the rupture of the bond with the Ni₃S₂ surface, and is endothermic, which correlates with the experimental data.

On the other hand, the diffusion of S₈ molecule on the crystal surface is evaluated differently. The diffusion barrier between equivalent positions 2, i.e. the paths 2→2d and 2→2b and from position 2 to position 1 (Figure 3) were calculated using the elastic band method (NEB). The estimated results are respectively 2.04 eV and 2.51 eV. The considered paths of sulfur atom movement allow us to conclude that the diffusion of sulfur atoms will be limited by the movement of sulfur from position 2 by high barriers of 2.49 eV and 2.04 eV, which will complicate the overall kinetics of sulfur diffusion along the surface of the Ni₃S₂ crystal.

Upscaling studies

To simulate the process of hydrogen sulfide decomposition in a non-stationary mode, a kinetic model was developed based on experimental data set obtained on laboratory stage. The rate of hydrogen sulfide decomposition is described by a differential equation:

$$\frac{d[H_2]}{d\tau} = f(TOS) = \frac{k_1 \cdot [H_2S] \cdot (\alpha \cdot [Ni_3S_2])^{1.5}}{1 + k_2 \cdot [H_2]}$$

where α is fraction of non-deactivated catalyst, $\alpha=1$ at TOS=0.

The overall process flow diagram (PFD) is provided in [Figure 4](#). The process of catalytic decomposition of hydrogen sulfide can be implemented using the following process stages:

1. Compression of the source gas to a pressure of 3 bar.

The necessity of this operation is due to the high hydraulic resistance of the catalyst layer and the entire chain of devices. In this case, a single-stage compressor can be used.

- ## 2. Catalytic decomposition of hydrogen sulfide.

This stage is carried out in two alternating reactors operating at a temperature of 525 °C. The feed flow is partially heated to the reaction temperature by a recuperative heat exchanger and is further heated by an electric heater or a furnace. With a hydrogen sulfide concentration in the flow of 10 vol %, the adiabatic heating will be about 15 °C, which allows the process to be carried out in an adiabatic mode without removing heat from the reaction zone.

- ### 3. The catalyst regeneration

The procedure is carried out by blowing hot nitrogen (or any gas inert under given conditions, including CO₂) at a temperature of 650 °C. Under these conditions, the excess sulfur accumulated on the catalyst during reaction is removed from the catalyst structure, evaporated and carried away from the reactor. Afterwards, sulfur is separated in a separator at a temperature of 100 °C. Nitrogen (or inert gas including CO₂) from the separator circulated back to the system using a circulation compressor.

4. Separation of unreacted hydrogen sulfide by physical absorption with methanol (or amines).

The stream containing unreacted hydrogen sulfide and hydrogen from the reactors is sent to an absorption column to separate hydrogen sulfide. The purified gas is sent to the next stage, and the saturated methanol solution is fed to a regeneration column, from where it is then recycled.

- ## 5. Product purification

Removal of hydrogen sulfide residues by hot chemisorption on zinc oxide is necessary for purifying the hydrogen-rich gas mixture and feeding it to the pressure swing adsorption (PSA) unit. The process is carried out in a flow-through adsorber with a fixed sorbent layer at a temperature of 450 °C.

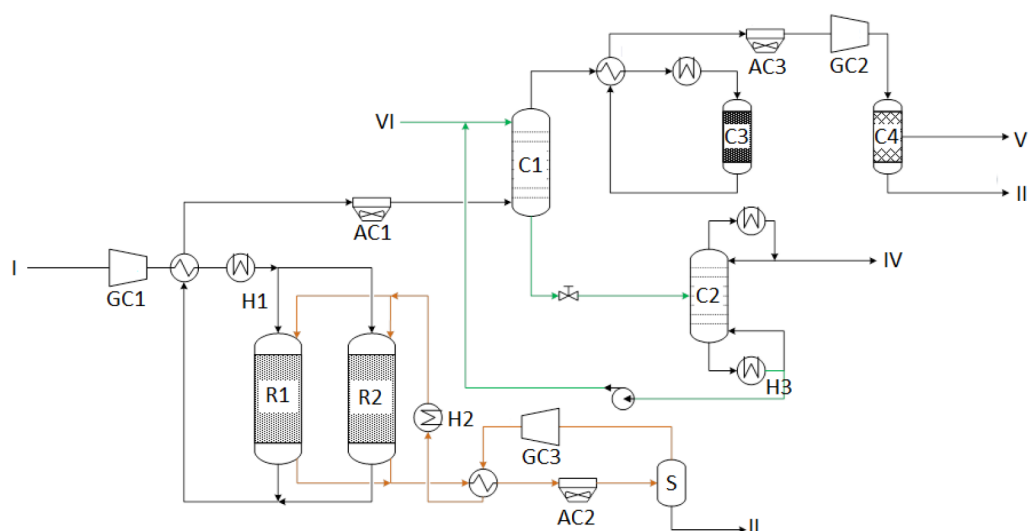


Figure 4—Technological scheme of the H₂S to H₂ process: GC1...3 – compressors, H1...3 – heaters, AC1...3 – air coolers; R1,2 – flow-through reactors with catalyst, S – sulfur separator, C1,2 – H₂S separation units, C3,4 – H₂ purification units; I – H₂S enriched feed gas, II – solid sulfur, III – pure H₂ gas, IV – separated sour gas, V – impurities separated from H₂.

To obtain commercial grade hydrogen applicable for fuel cells, additional PSA adsorption stage is required. PSA using molecular sieves has greater selectivity compared to membrane separation, but requires compression of the flow to a pressure of 2.0 MPa.

This process was modeled using conventional chemical engineering process simulator. The asymmetric NRTL method was used to calculate the thermodynamic properties of the system. The hydrogen sulfide decomposition reactors were modeled using conversion reactors based on the results of kinetic calculations. The hydraulic resistance in the catalyst bed was assumed to be 0.1 MPa. In other devices, the hydraulic resistance was neglected. The efficiency of the pumping equipment was estimated using the internal database of the used software in the range of 26-30%.

When modeling the absorption-regeneration unit of hydrogen sulfide with methanol, the absorption column was operated on the basis of reaching the concentration of residual H_2S no more than 100 ppm. The absorbent regeneration column was calculated based on achieving the degree of H_2S extraction of 99%, and the concentration of methanol in the gas blow-off from the column of no more than 100 ppm.

In order to increase the hydrogen yield and reduce the specific energy costs of the scheme, the process flow diagram was optimized. The pressure of the initial flow was increased to 2.0 MPa by creating an additional compression stage in the compressor. This was done to increase the efficiency of the methanol absorption column, as well as to remove the compressor from the diagram before the PSA unit. Also, the flow of hydrogen sulfide separated in the regeneration column was sent for recycling to the second stage of the injection compressor. When adding a circulation loop, the concentration of hydrogen sulfide in the flow before the reactor increases to 22%, provided that a fixed conversion value of 38% is achieved. According to the results obtained, optimization of the scheme led to a significant increase in hydrogen yield and a decrease in specific energy costs by almost 6 times.

Conclusions

Through meticulous screening of potential catalysts and process parameter optimization, we were able to obtain efficiency that exceeded findings previously reported by other researchers. Specifically, our selected catalyst demonstrated the ability to convert 40% of the H_2S in the flow to hydrogen at a relatively low temperature of 500 °C. In this case, the catalyst serves a dual purpose by simultaneously separating sulfur and hydrogen, thereby preventing the reverse formation of H_2S . The catalyst structure securely stores sulfur during the reaction, and it can purge it during regeneration by simply raising the temperature. Through extensive lab-scale test runs, we accumulated the necessary data to develop a process flow diagram including pretreatment, reaction, and hydrogen separation phases. By using technological design optimization and computational simulations, we were able to decrease process energy consumption by a factor of 6. This data appears sufficient to proceed with the technology's piloting. We conclude that the incorporation of this technology into existing gas plants can support the hydrogen economy by generating hydrogen from sour gas.

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